SPACE X

Predicting Falcon 9 First-Stage Landings

Applied Data Science Capstone • Technical Report

A reproducible pipeline from API and web data to interactive analytics and classification.

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https://github.com/ijontichyai/IBM-Data-Science-Capstone

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Executive summary

Can launch history reveal when a Falcon 9 first stage is likely to land?

90
launch records analyzed

76.9%
KSC LC-39A success rate

84.8%
best cross-validation score

83.3%
verified holdout accuracy

What we did

- Collected launch data via API and web scraping
- Cleaned, encoded, queried, mapped, and visualized the data
- Tuned Logistic Regression, SVM, Decision Tree, and KNN

What we learned

- Launch site, payload, orbit, and booster context matter
- SVM led validation; SVM and Logistic Regression tied on test data
- False positives—not missed landings—were the main error mode

Business context and analytical problem

WHY IT MATTERS

First-stage reuse can reduce launch cost and improve mission economics.

A reliable landing prediction helps planners estimate reuse potential before launch.

Research question

Given launch conditions and historical outcomes, will the first stage land successfully?

INPUTS	Payload • Orbit • Site • Booster • Flight history
TARGET	Class = 1 landed | Class = 0 did not land
OUTPUT	Transparent evidence for operational decision-making

From raw launch records to decision evidence

01
COLLECT
API + Wikipedia

02
WRANGLE
Normalize + label

03
EXPLORE
Plots + SQL

04
INTERACT
Folium + Dash

05
PREDICT
GridSearchCV

Design principle

Every result is traceable to source data, transformations, and a reproducible notebook or Python file.

Data collection — SpaceX REST API

GET
/v4/launches/past

LOOK UP
rockets · payloads · launchpads · cores

NORMALIZE
JSON → pandas DataFrame

FILTER
Falcon 9 launch records

EXPORT
structured CSV

Core fields retained

- Flight number and date
- Booster / core configuration
- Payload mass and orbit
- Launch site and landing outcome

Quality controls

- Resolve related IDs before analysis
- Remove empty or irrelevant records
- Preserve a repeatable extraction path
- Store the final analytical table

Data collection — web scraping

1. Request Wikipedia launch-history page
2. Parse HTML tables with BeautifulSoup
3. Extract rows and normalize cell text
4. Build a structured launch-record DataFrame
5. Validate columns and export for analysis

Fields extracted

- Flight No.
- Launch site
- Payload / orbit
- Customer
- Mission outcome
- Booster landing

Data wrangling and feature engineering

CLEAN

Handle missing payloads, normalize text, remove duplicates

LABEL

Convert landing outcome to binary class 0 / 1

ENCODE

One-hot encode orbit, site, landing pad, and serial

SCALE

Standardize numeric features before distance-based models

Result: one analysis-ready feature matrix X and one target vector Y.

Exploratory analysis — questions before models

SITE

Where do launches succeed most often?

PAYLOAD

How does mass relate to outcome?

ORBIT

Which mission profiles are most successful?

BOOSTER

Which configurations perform best?

TIME

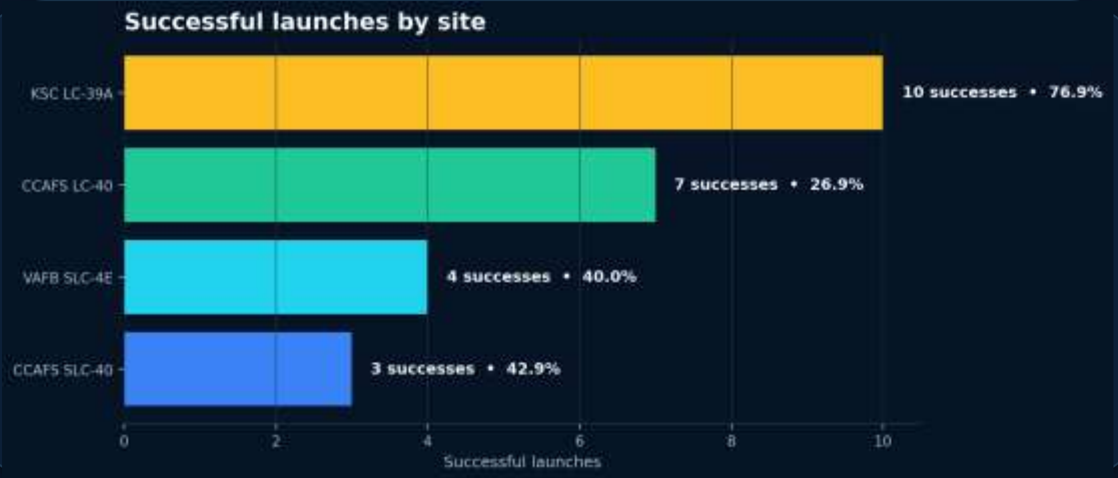
Does success improve as experience grows?

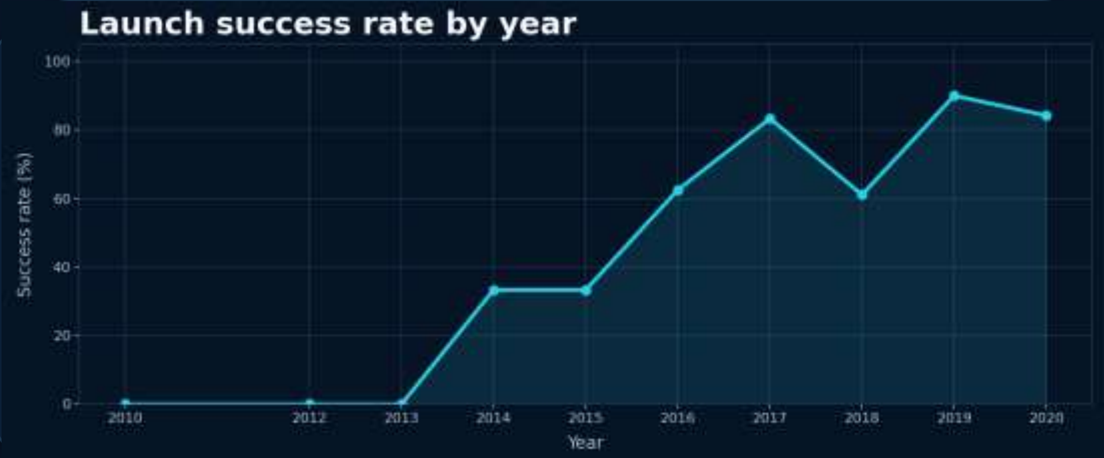
RISK

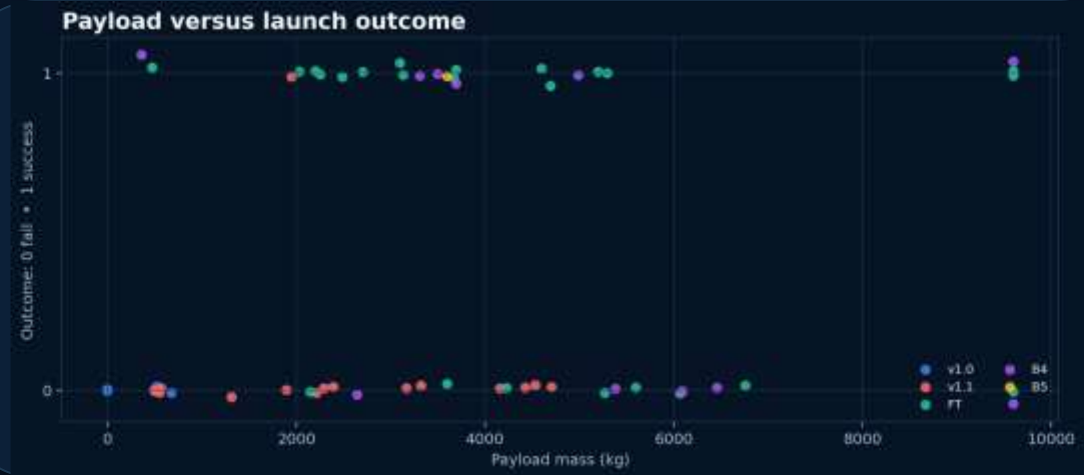
Where do failures cluster?

Charts explain relationships; SQL verifies exact aggregations.

EDA results — site, payload, and experience







EDA with SQL — reproducible analytical questions

01 **SITE COVERAGE** — DISTINCT launch sites and CCA-prefixed records

02 **PAYLOAD** — SUM and AVG payload mass by customer / booster

03 **SUCCESS** — COUNT outcomes and drone-ship landings

04 **RANKING** — GROUP BY booster and ORDER BY success count

05 **TIME** — Filter launch dates and compare historical periods

Pattern: SELECT → WHERE → GROUP BY → aggregate → ORDER BY

SQL results — exact summaries support the story

QUESTION	RESULT	USE
Launch sites	Four sites represented in the analytical dashboard	Coverage
Payloads	Observed payload range extends to about 9,600 kg	Distribution
Success rates	KSC LC-39A: 76.9% • CCAFS LC-40: 73.1%	Performance
Ranking	KSC LC-39A contributes the largest share of successes: 41.7%	Ranking
Time analysis	Later launch sequences show stronger and more stable outcomes	Trend

Interactive geospatial analytics with Folium

MAP LAYERS

1	**Launch sites**	Circle markers + labels
2	**Launch records**	Green success / red failure markers
3	**Clusters**	MarkerCluster for overlapping launches
4	**Proximity**	MousePosition + distance lines

PROXIMITY TESTS

- **Coastline**
- **Road network**
- **Railway**
- **City / population center**

Goal: explain why launch facilities occupy their specific geography.

Folium results — coastal sites and controlled access

Key geographic findings

- All sites are positioned near a coastline
- Transport infrastructure supports logistics
- Launch areas remain separated from dense cities
- Clusters reveal repeated operations at the same pads

Operational logic: access + safety + ocean corridor

Plotly Dash — turning analysis into an interface

INPUTS

Launch-site dropdown

Payload range slider

USER ACTION

Select a site and constrain payload mass.

CALLBACKS

Input values trigger filters on the shared DataFrame.

Output(..., "figure") routes each result to the matching graph ID.

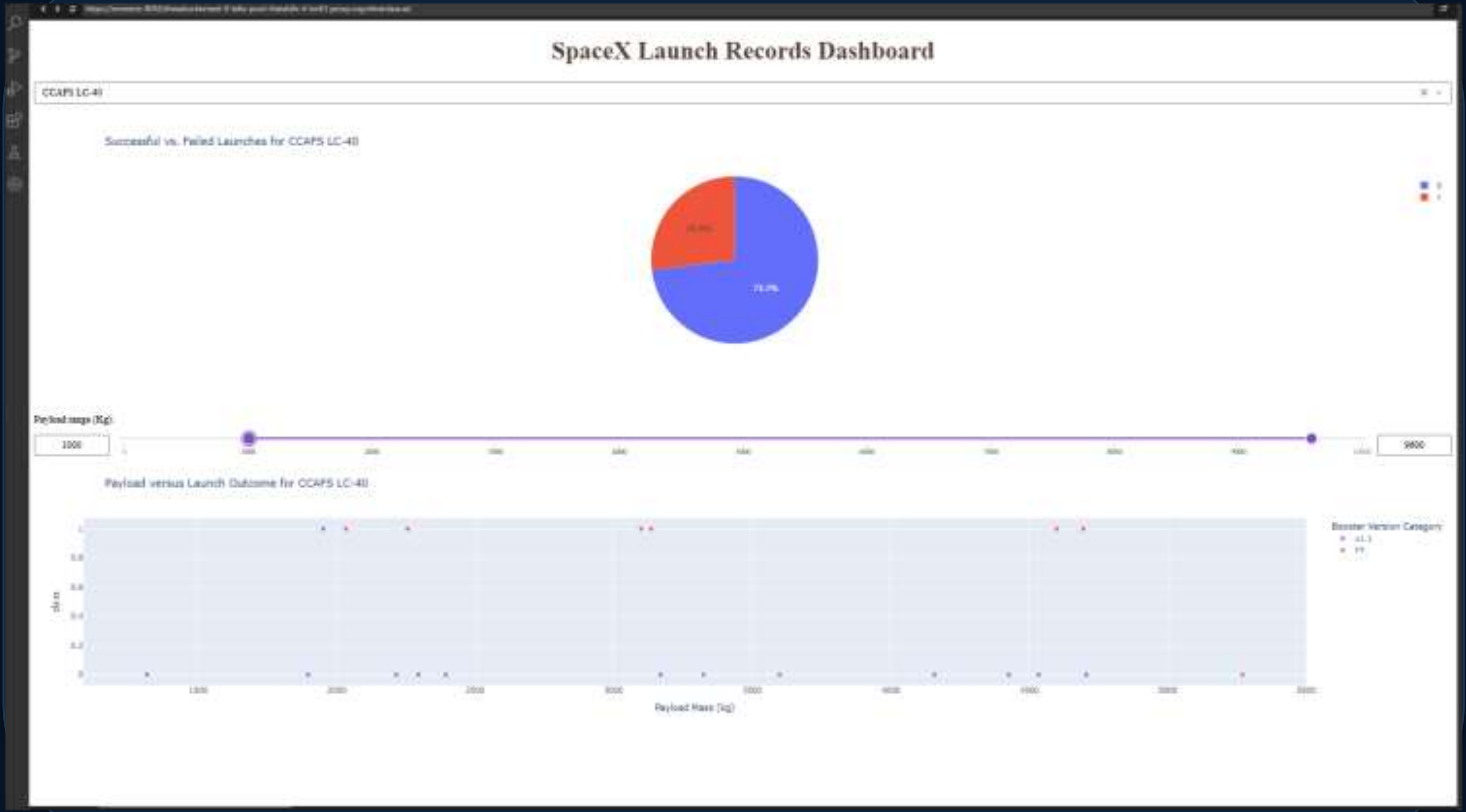
OUTPUTS

Success pie chart

Payload scatter plot

Interactive evidence updates in real time.

Dash results — site and payload filters work together



Observed insights

41.7%

share of all successes at KSC LC-39A

76.9%

KSC LC-39A success rate

The payload slider removes out-of-range missions; the site dropdown updates both visuals through callbacks.

Predictive analysis — fair model comparison

SPLIT
80% train / 20% test
random_state = 2

STANDARDIZE
Fit scaling on X
reassign transformed X

TUNE
GridSearchCV
10-fold CV

EVALUATE
Holdout accuracy
confusion matrix

Models evaluated

- **Logistic Regression** — C · penalty · solver
- **Support Vector Machine** — C · gamma · kernel
- **Decision Tree** — depth · split · features
- **K-Nearest Neighbors** — neighbors · algorithm · p

Model results — validation lead, test tie, clear error mode

Verified accuracy

Logistic Regression

- CV 84.6%
- Test 83.3%

SVM — sigmoid

- CV 84.8%
- Test 83.3%

SVM has the strongest validation score; both verified models classify 15 of 18 test records correctly.

Confusion matrix

	Predicted fail	Predicted land
Actual fail	3	3
Actual land	0	12

3

false positives

0

false negatives

Main risk: predicting a landing when the stage actually fails.

Conclusions and innovative next steps

What the evidence supports

- KSC LC-39A leads both success volume and site success rate
- Launch outcomes improve across the flight sequence
- SVM gives the best validation score; Logistic Regression matches its test accuracy
- False positives deserve the most operational attention

Beyond the template

1. **COST-AWARE THRESHOLD**

   Penalize false positives more heavily than false negatives.

2. **CALIBRATED PROBABILITY**

   Report landing probability—not only a binary class.

3. **DRIFT MONITORING**

   Retrain as new booster versions and mission profiles appear.

Recommendation: deploy as decision support, not an autonomous go/no-go rule.